Research project overview

standai: a 2D humanoid that learns to get up

One neural network trained with PPO · Nov 2026

standai overview and key results

A single PPO-trained network gets up from lying down and is still standing 20 s later in 94.2% of starts

94.2%

Lying start → standing 20 s later

Mean of supine / prone, 50:50

600 starts per side, unseen seeds

97.7% · 90.7%

Success by start side

Supine (on the back) 97.7%

Prone (face down) 90.7%

+23.4 pt

vs. rule-based controller

70.8% → 94.2%

0/160 at the start of the project

Task	A 2D MuJoCo ragdoll gets up from lying on its back or face down and stays standing
Method	Train role-specific policies separately → distil into one network (DAgger) → end-to-end PPO fine-tuning
Measurement	Simulation only; fixed seeds, deterministic policy; ±2.5 pt sampling error

※ Verified in 2D simulation only; start poses come from the environment's own generator; one training run per configuration

Problem: why is getting up from the floor hard?

A single policy never got up from the floor (0/160), and one policy could not learn floor motion and balance together

Task definition

Start

Lying on the back (supine) or face down (prone)

Success

Still standing 20 s later

Evaluation

Fixed seeds, deterministic policy, 600 starts per side

What made it hard

A single policy could not get up

Reward shaping, replaying failed starts, a 1.5× stronger body, exploration noise: none changed the 0/160 on lying starts

Floor motion and balance conflict

The policy alone raised the head above 0.5 m in only 4–8% of floor starts (random patterns: 77%). One policy's action noise cannot be wide on the floor and narrow while balancing

Training logs hide real progress

Log metrics sometimes moved opposite to real success, and the same seeds shift ±4 pt between runs, so everything was re-measured on fixed seeds

Approach: grow the pieces, distil, fine-tune end to end

Grow role-specific policies first, merge them into one network, then fine-tune it end to end

Grow the pieces

Train the floor (supine / prone), hand-over and stand policies separately, each with its own reward and curriculum, joined by a rule-based controller

Standing after 20 s
24% → ~72%

Distil into one network

Distil the whole controller, switching rules included, into one 512×512 network with DAgger

Standing after 20 s
74.5% (rules: 73.8%)

End-to-end fine-tuning

Fine-tune with PPO on a 20 s 'standing' reward for 40M steps (value head warmed up first)

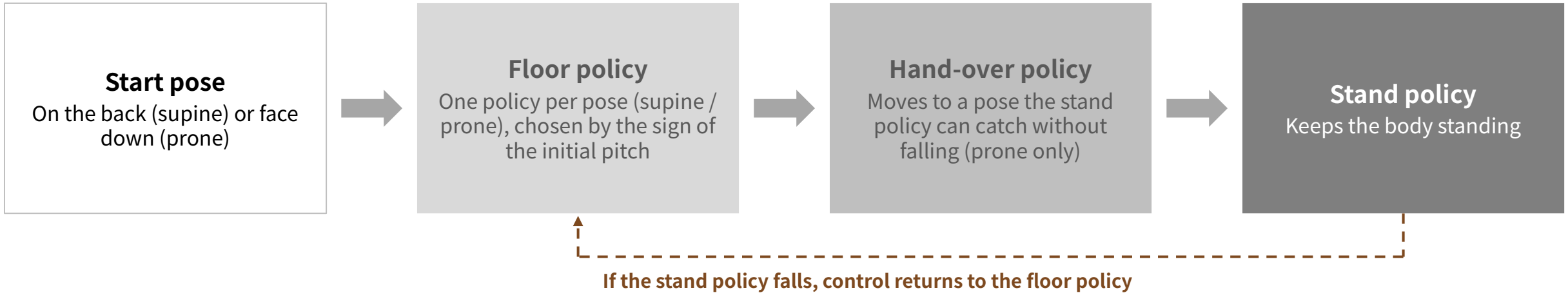
Standing after 20 s
94.2%

Big networks trained from scratch never got off the floor → growing the pieces first is the key

※ The 24% → 72% of the first step is the several-week improvement of the rule-based controller (see the progress slide)

Controller: three policies joined by rules

The start pose (sign of the initial pitch) picks the floor policy, which hands over to the stand policy via a hand-over policy



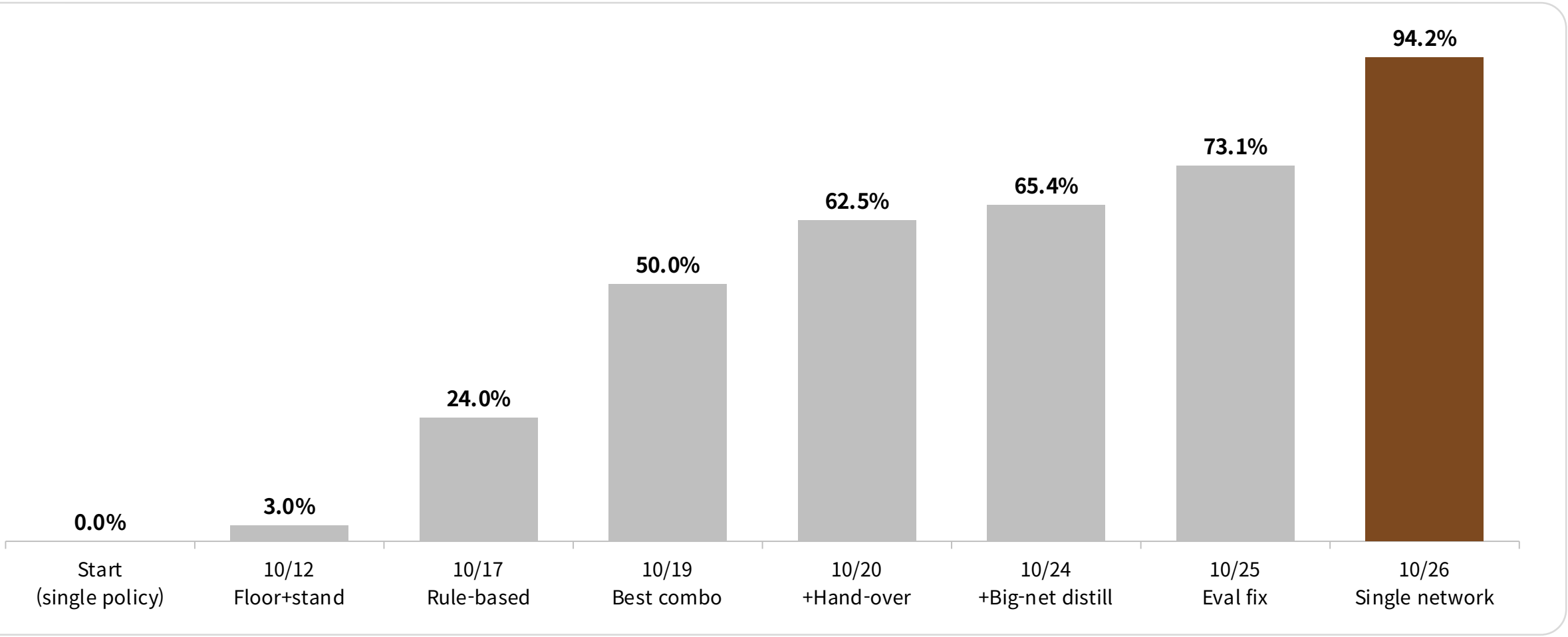
Final rule-based controller (standing after 20 s)

73.1%	61.8%	84.4%
50:50 average	Supine	Prone

※ After fixing an evaluation bug; six fresh seed sets (1,200 starts per side). Most losses came at the floor → stand seam; a single network has no seam, which is the presumed reason for the gain

Progress: from 0% to 94%

The last jump came from training one network end to end (73% → 94%)

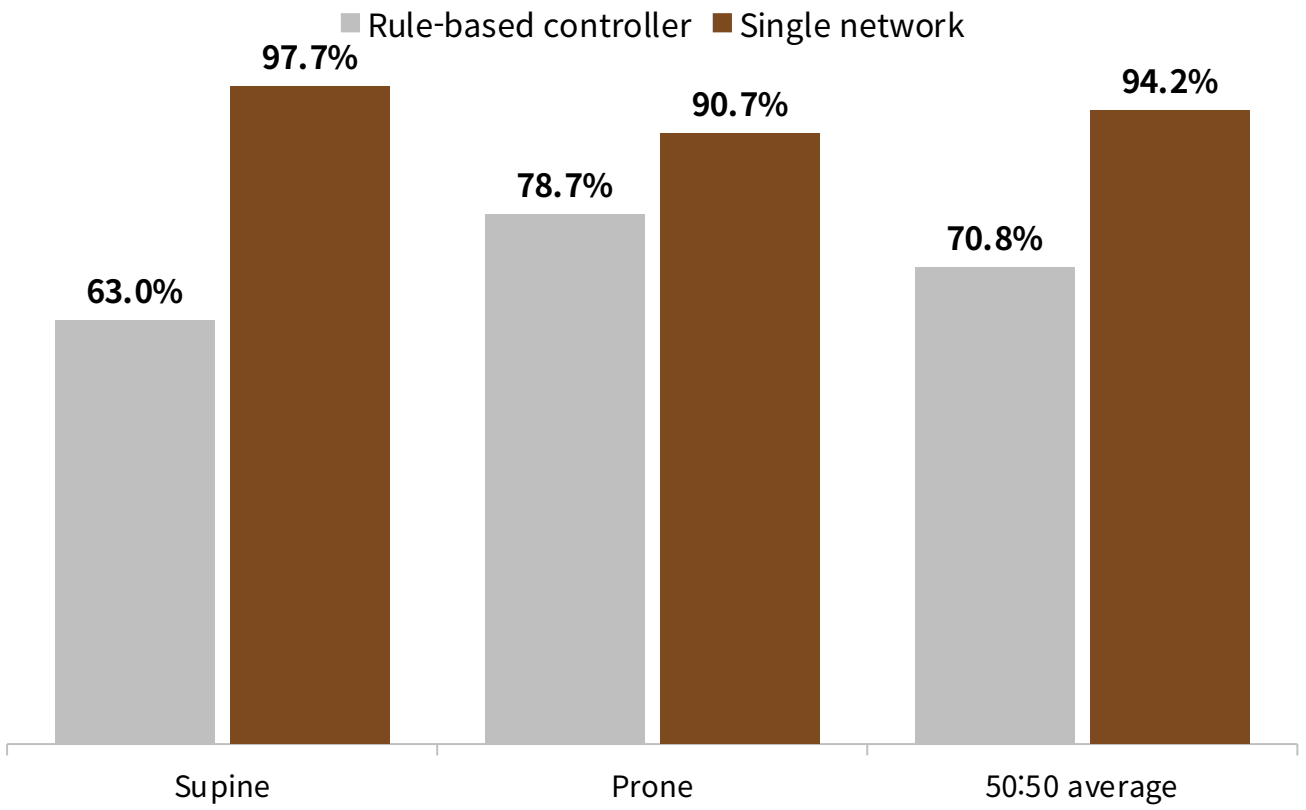


※ Standing after 20 s (%). Test sets, start mixes (80:20 → 50:50) and seeds differ between stages, so this is not a strict comparison. Start: 0/160 for the single policy; 10/12: 3/100

Final result: one network beats the rule-based controller

On the same seeds: 94.2% vs. 70.8% for the rule-based controller, with large gains on both start sides

Standing after 20 s



Checks

Test setup

Fresh seeds (99100 · 99200 · 99300), 600 starts per side, ±2.5 pt sampling error

Shortcut checks

No unphysical force (root_x, root_ry) is applied; moving the opponent far away changes nothing; pushes of 150–500 N after the get-up change nothing

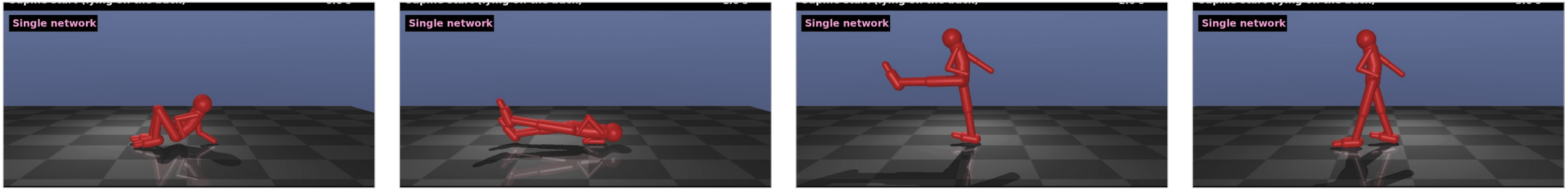
Calm standing

Torso-height std 0.009 m, joint speed 0.17–0.46 (rules: 1–2)

Demo: getting up from lying down

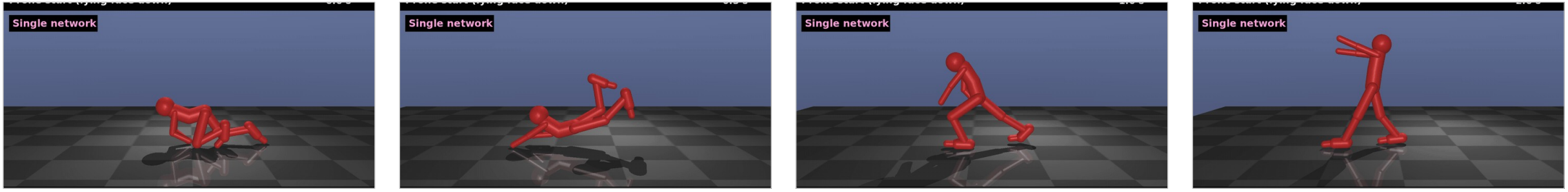
Two selected runs (supine / prone): standing within about 2.5 s

Start on the back (supine)



0.0 s　1.0 s　2.0 s　3.0 s

Start face down (prone)

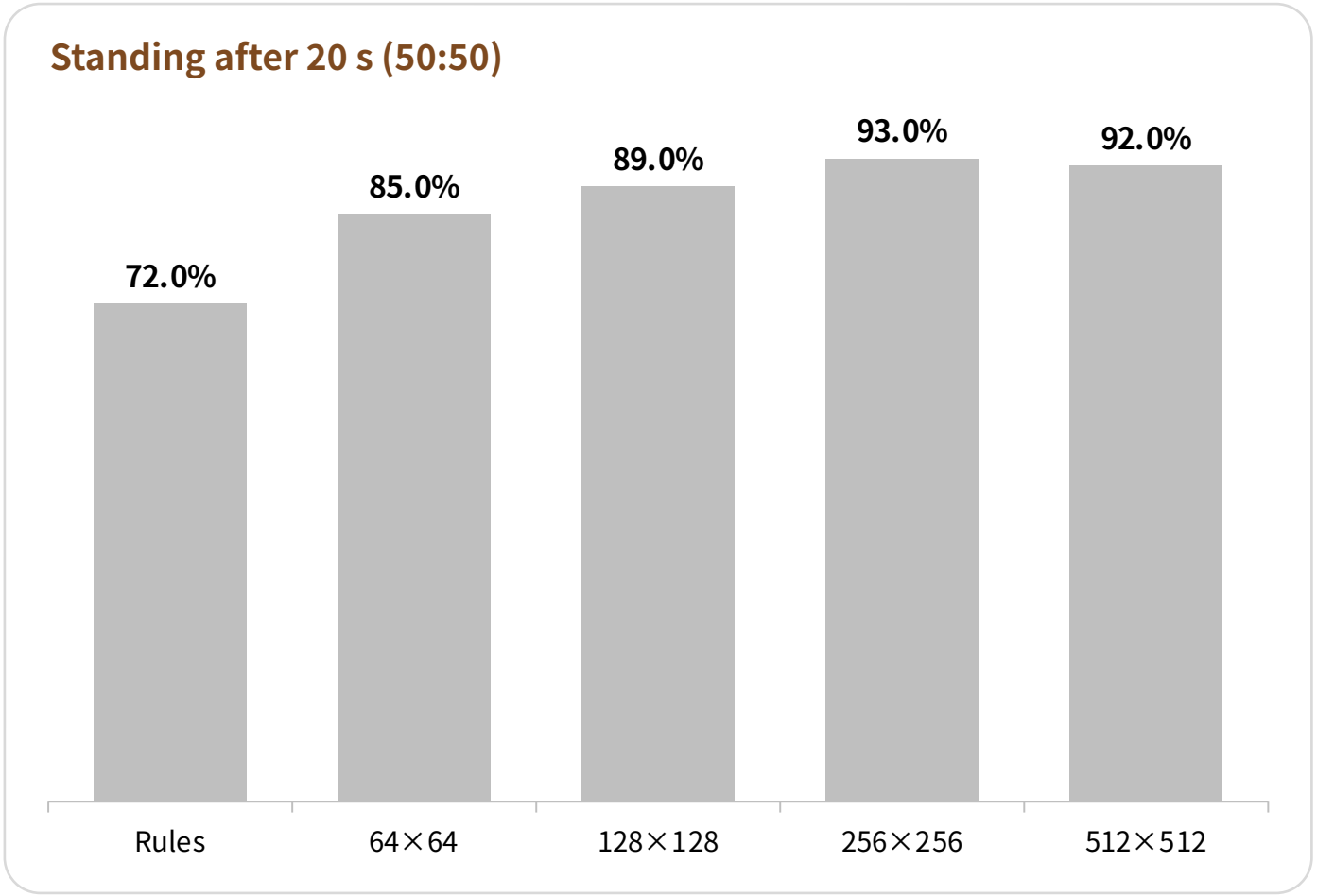


0.0 s　0.5 s　1.0 s　2.0 s

※ Selected runs, not typical: about 98% (supine) and 91% (prone) of starts end up standing after 20 s. Source: standai/assets/demo.gif

Network size was not the bottleneck

Shrinking to 256×256 keeps performance; even 64×64 beats the rules by 13 pt → the recipe made the gain



How to read

Capacity is not the bottleneck

Performance holds down to 256×256 (93%); 128×128 gives 89%

Small still beats rules

64×64 reaches 85% vs. 72% for the rule-based controller

Big nets from scratch fail

Big networks trained from scratch on the floor task failed repeatedly → grow the pieces, distil, fine-tune

※ Same recipe (distillation, then 12M end-to-end steps), fresh seeds 99500+, 200 starts per side. One run per size, so differences of ~3 pt are not resolved. A big network trained from scratch was not run in this comparison

Lessons: what worked and what did not

Growing small policies, distilling and lowering the learning rate worked; stacking rewards mostly did not

What worked

Distil, then fine-tune

Distil a working small policy into a bigger net, then fine-tune end to end (big nets from scratch fail)

Lower learning rate

Stand-policy PPO 3e-4 → 5e-5: standing after 20 s 29% → 41%

Arrival-quality reward

Reward the floor policy for arriving in a state the stand policy can catch; supine 49.5% → 60%

What did not work · caveats

Big nets from scratch

A floor policy trained from scratch as a big network (256×256) kept failing; distilling a small policy worked

Stacking rewards and timing

Reward stacking, re-training the stand policy, hand-over timing gates and critic-based rewards did not move the rule-based system

Evaluation bugs flip conclusions

A hand-over policy applied to the wrong start side under-reported the result (68%) → 73.1% after the fix; new measurements are checked against the official controller

Limits and roadmap

Results are simulation only; next steps are standing from any posture → walking → transplant into combat

Limits

Scope
Simulation and 2D only; start poses come from the environment's own generator (others untested)

Statistics
One training run per configuration, ±2.5 pt sampling error, same seeds move ±4 pt between runs

Assumption
The 50:50 supine/prone mix is an assumption; the real fall-direction mix is unknown

Scale
About 1B steps in total (get-up papers use ~5B); limb torque ×1.5 without scaling link masses

Roadmap

1. Standing from any posture

A policy that stands from any pose, not only lying

2. Walking

Extend to velocity tracking

3. Transplant into combat

Move into fightai's combat action/observation space (transplant)

Related work: multi-expert distillation (Rusu 2015, Actor-Mimic 2016), DAgger (Ross 2011), growing local policies then distilling (Ghosh 2018), getting up (Tao 2022, He 2025). Not a new technique: the contribution is an experimental record of how much each step mattered on this task